

Practical 4 - Cox regression

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `pbcbase.dta`, `alloauto.dta`

Introduction

This practical is in two parts.

Part A. We will fit Cox models to the PBC data used in Practicals 1 and 2, assess the proportional hazards assumption, and estimate survival curves for different values of the covariates.

Part B. We will use a new dataset, called `alloauto`, to investigate the proportional hazards assumption of the Cox model.

Aims

The aims of this session are:

- Fit Cox proportional hazards models and interpret the results.
- Obtain estimates of hazard ratios to compare hazards for individuals with different values of covariates.
- Obtain estimates of survival curves from a Cox regression model for particular values of the covariates.
- Use methods to assess the proportional hazards assumption.

Part A: Primary Biliary Cirrhosis (PBC) data

We will start by estimating the association between treatment group (**treat**) and the hazard for mortality, using Cox regression. Later in the practical we will also use the **bil0** variable. The time origin is the time of randomisation. The variables of interest for this practical are described below.

Variable	Description
id	Unique identifier for each participant
datein	Date person entered the study
dateout	Date of the end of follow-up due to either death or censoring
d	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
time	Follow-up time in years
treat	Treatment: 1=placebo, 2=active
bil0	Serum bilirubin (mg/dl), measured at the start of the trial.

Load the data using:

```
use "pbcbase.dta" , clear
```

Question 1

- Write down the form of the hazard assuming a Cox proportional hazards model for a single covariate X (representing treatment).
- Write down the partial likelihood that can be used to estimate the parameter of this model.

In Cox regression, there is a contribution to the partial likelihood from each event time. In this question we will derive the contribution to the partial likelihood at time $t = 0.052$ in the PBC data.

- What is the value of **treat** for the individual who has the event at time $t = 0.052$?
- How many individuals are at risk at time $t = 0.052$?
- What values of **treat** do these individuals have?
- Using the above information find the contribution to the partial likelihood at time $t = 0.052$.

The output from the following will help:

```

** Set up the data for survival analysis: time in study timescale
stset time, failure(d)

** Take a look at the first few lines of the Kaplan-Meier table
sts list if _t<0.2, by(treat)

** Look at people who were in the study until at least t=0.052
tab treat if time>=0.052

```

SOLUTION

- The hazard is $h_0(t)$ in the control group and $h_0(t)e^\beta$ in the treatment group. We can write the hazard as $h(t) = h_0(t)e^{\beta X}$ where $X = 0$ denotes the control group and $X = 1$ denotes the treatment group.
- The partial likelihood is

$$L_P = \prod_j \frac{\exp(\beta x_{i_j})}{\sum_{k \in R_j} \exp(\beta x_k)}$$

where i_j is the index of the person who has the event at time t_j , R_j denotes the risk set at time t_j , and the product is over all event times t_j ($j = 1 \dots, J$).

- The person who had the event at time $t = 0.052$ was in the placebo group. Hence $X = 0$.
- There were 181 people at risk at time $t = 0.052$. This is the risk set.
- The risk set at time $t = 0.052$ includes 89 people in treatment group 1 (placebo, $X = 0$) and 92 in treatment group 2 (active, $X = 1$).
- The contribution to the partial likelihood at event time 0.052 is therefore

$$\frac{e^{\beta \times 0}}{89e^{\beta \times 0} + 92e^{\beta \times 1}} = \frac{1}{89 + 92e^\beta}$$

Question 2

Fit the Cox model including treatment as the covariate using the following code, and interpret the results:

```
stcox i.treat
```

SOLUTION

- The estimated hazard ratio comparing those in the treatment group relative to the control group is 0.86 - that is, the hazard in the active treatment group is estimated to be 14% lower than in the placebo group.

- However, the confidence interval for the estimate is fairly wide and includes the value 1, indicating no evidence against the null hypothesis that the hazard rate for death is the same in the two treatment groups, which is the hypothesis that $\beta = 0$.
- The p-value from a Wald test of this null-hypothesis is 0.45. Note that a likelihood-ratio test could alternatively have been used – the conclusion would be the same.

Question 3

- Obtain the estimated survival curves in the two treatment groups based on the Cox model using the code below.

```
stcurve, survival at1(treat=1) at2(treat=2)
```

- What is the probability of survival beyond time 5 in the two treatment groups? You can read these roughly from the plot. See the do file for how to get the estimates (there is not a nice way to do this in Stata).
- Why do the estimated survival curves have ‘steps’?
- Why do these survival curves differ from the Kaplan-Meier estimates?

SOLUTION

- The survival curves show that estimated survival is slightly higher in the active treatment group throughout the study. In the placebo group 1 the probability of survival beyond time 5 is 0.517 (95% CI 0.423 to 0.632). In the active treatment group the probability of survival beyond time 5 is 0.569 (95% CI 0.474 to 0.682).
- The survival curves obtained after fitting the Cox model have ‘steps’ due to the semi-parametric nature of the Cox model - we have not specified a shape for the baseline hazard.
- The Kaplan-Meier survival curves are non-parametric, meaning they do not assume anything about the shape of the hazard or about the relationships between the hazards in the two treatment groups. The survival curves obtained after fitting the Cox model make use of the hazard ratio and the estimated cumulative hazard – to get these curves we have made an assumption about the relationship between the hazards in the two treatment groups. Specifically, we have made the proportional hazards assumption.

Question 4

- Assess the proportional hazards assumption graphically. You can make use of a non-parametric plot in the same way as you did to assess a Weibull model in Practical 3. See the code below.
- What do you conclude about the proportional hazards assumption for these data?

```
stphplot, by(treat)
stcoxkm, by(treat)
```

SOLUTION

- We can assess the proportional hazards assumption for a binary variable by looking at a plot of $\log\{-\log S(t|x)\}$ (estimated using Kaplan-Meier) against $\log t$ in the two treatment groups ($X = 0, 1$). If the proportional hazards assumption is valid, we expect the lines for the two groups to be approximately parallel over time.
- In this case the lines are very close together over time. Although they are crossing over each other in places, this is inevitable if they are so close together. There is little to suggest the proportional hazards assumption is not valid based on this plot.
- The second plot compares the estimated survival curves from the Cox model with the Kaplan-Meier survival curve estimates. If the proportional hazards assumption is met we expect the two sets of curves to be similar, which they are. This plot provides no evidence to suggest that the proportional hazards assumptions not valid.

Question 5

Next we investigate how the level of bilirubin measured at the start of the trial (bil0) is associated with the outcome.

- Write down the form of a hazard model including both treatment group and baseline bilirubin.
- Fit the model and interpret the results.

SOLUTION

- Let X_1 denote treatment status now ($X_1 = 0$ for control group and $X_1 = 1$ for the treatment group), and let X_2 denote bilirubin measured at baseline. The hazard model is

$$h(t|X_1, X_2) = h_0(t)e^{\beta_1 X_1 + \beta_2 X_2}$$

where $h_0(t)$ is the hazard for a person in the placebo group with baseline bilirubin level of 0, β_1 is the log hazard ratio associated with treatment after adjusting for bilirubin, and β_2 is the log hazard ratio for bilirubin after adjusting for treatment.

- After adjusting for baseline bilirubin, the HR for treatment versus control is 0.68 (95% CI 0.45, 1.04), and after adjustment for treatment group the HR for a 1 unit increase in baseline bilirubin is 1.007 (95% CI 1.006, 1.009). After adjustment for bilirubin, treatment is associated with a 32% reduction in the hazard, and there is some borderline evidence ($p=0.073$) that the hazard ratio differs from 1. After adjustment for treatment,

a 1 unit increase in bilirubin is associated with a 0.7% increase in the hazard, and this association is highly statistically significant.

- Looking at the distribution of bilirubin (e.g. in a histogram) we see that a 1 unit change is unlikely to be clinically meaningful, since the mean bilirubin is 60 and the standard deviation is 80. So it may be sensible to change the units for bilirubin. From the above results we can calculate that a 50 unit increase in bilirubin is associated with a 43% increase in the hazard ($1.007^{50} = 1.43$) after adjusting for treatment group, with 95% CI 1.31 to 1.55.

Question 6

We will now compare the hazards in different types of individual, based on the treatment and baseline bilirubin value.

- What is the hazard ratio comparing: (i) a person in the active treatment group with `bil0=75`, (ii) a person in the active treatment group with `bil0=30`.
- What is the hazard ratio comparing (i) a person in the placebo group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.
- What is the hazard ratio comparing (i) a person in the active treatment group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.

SOLUTION

- What is the hazard ratio comparing: (i) a person in the active treatment group with `bil0=75`, (ii) a person in the active treatment group with `bil0=30`.

$$\frac{h_0(t) \exp(\beta_1 \times 1 + \beta_2 \times 75)}{h_0(t) \exp(\beta_1 \times 1 + \beta_2 \times 30)} = \exp(\beta_2 \times 45)$$

- The estimated HR is 1.38 (95% CI 1.28 to 1.48).

- What is the hazard ratio comparing (i) a person in the placebo group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.

$$\frac{h_0(t) \exp(\beta_1 \times 0 + \beta_2 \times 75)}{h_0(t) \exp(\beta_1 \times 0 + \beta_2 \times 30)} = \exp(\beta_2 \times 45)$$

- The estimated HR is 1.38 (95% CI 1.28 to 1.48) – as in (a).

- What is the hazard ratio comparing (i) a person in the active treatment group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.

$$\frac{h_0(t) \exp(\beta_1 \times 1 + \beta_2 \times 75)}{h_0(t) \exp(\beta_1 \times 0 + \beta_2 \times 30)} = \exp(\beta_1 + \beta_2 \times 45)$$

- The estimated HR is 0.94 (95% CI 0.63 to 1.41).

Question 7

- Obtain the estimated survival curves for individuals in the two treatment groups with baseline bilirubin value equal to 15, 30 and 75 (these are approximately the 25th, 50th and 75th percentiles). You can do this by extending the code used in question 4.
- What is the estimated probability of survival beyond 5 years for these six example individuals?

SOLUTION

- Those in the control group and with high baseline bilirubin have the worst survival, while those in the treatment group with low baseline bilirubin have the best survival.
- The estimated survival probabilities at 5 years range from 0.439 for individual 5 (`treat=1,bil0=75`) to 0.693 for individual 2 (`treat=2,bil0=15`).

Question 8

Fit a Weibull model including treatment and bilirubin levels as covariates. Compare your results from the Cox model with those from a Weibull model.

SOLUTION

The hazard ratio estimates from fitting a Weibull model with treatment and baseline bilirubin as covariates are almost identical to those obtained using the Cox model. However, by using the Cox model we avoided having to make an assumption about the shape of the baseline hazard.

Part B: Bone marrow transplant data

The `alloauto` dataset is from a study of 101 individuals with advanced acute myelogenous leukemia. 51 of the patients received treatment using their own bone marrow (an autologous bone marrow transplant) and 50 patients received bone marrow from a sibling (an allogenic bone marrow transplant). The event of interest was a composite of death or relapse. The three variables in the dataset are summarised in the table.

Variable	Description
<code>time</code>	Time in months to event or censoring
<code>delta</code>	Event indicator: 0=Censored, 1=Death or relapse
<code>type</code>	Treatment type: 1=allogenic, 2=autologous

Question 1

Load the data and use `stset`. Obtain Kaplan-Meier estimates of the survival curves in the two treatment groups and perform a log rank test. Interpret your results.

SOLUTION

The Kaplan-Meier estimates of the survival curves in the two groups are close together just after the time origin, then they separate, before coming together again and eventually crossing over around time 18. The autologous group (type=2) have better survival up until around time 18, after which the allogenic group have better survival. There is substantial overlap in the confidence intervals. The result from a log rank test provides no evidence ($p=0.5$) of a difference between the two survival curves. The log rank test is valid but lacks power when survival curves are crossing.

Question 2

- Fit a Cox model including treatment type and interpret the results.
- Use graphical methods to investigate whether the proportional hazards assumption is appropriate for treatment type.

SOLUTION

The hazard ratio estimate is 1.19, indicating a higher hazard in the autologous group. The 95% CI is wide (0.68,2.09). The point estimate doesn't reflect the fact that we see better survival in the autologous group in the initial period of follow-up, and then worse survival in the later period. In this setting the hazard ratio will also differ depending on the length of follow-up, which is not a desirable feature.

The plot of $\log\{-\log S(t|x)\}$ against $\log t$ in the two treatment groups will show parallel curves if the proportional hazards assumption is valid. The fact that they are crossing here indicates evidence against proportional hazards, i.e. evidence that the ratio of the hazards between the two treatment groups is not constant over time.